

Particulate Matter Monitoring & Health Effects

Issue 2026-08-25 | Entry window 25 Aug 2026 | 14 in scope, 51 screened out

14RECORDS IN SCOPE
(51 SCREENED OUT)**7**OF 10 CLUSTERS
OCCUPIED**5**PREPRINT-STAGE
RECORDS (36%)**4**INTERVAL-BEARING
ESTIMATES (1 STUDY)**1**ABSTRACT-FREE
DEPOSIT (7%)

Signal of the day

A full year of riverbank co-location shows that the binding constraint on plume-resolved low-cost sensing is not correlation with the reference — it is noise floor and response time.

Eger et al. (*Aerosol Air Qual Res*) co-located a panel of low- and mid-cost sensors against reference instrumentation on the Rhine for one year, screening them specifically for the ability to resolve individual inland-ship exhaust plumes. Correlation was never the problem: **MA350 black carbon reached $R^2 = 0.95$ (slope 0.77), Partector 2 Pro ultrafine particle number $R^2 = 0.85$, K30 FR CO₂ $R^2 = 0.98$, and the Alphasense NO and NO₂ cells $R^2 = 0.99$ and 0.81.** What limited the study was that the sensors' instrumental noise and the temperature dependence of the electrochemical cells reduced the number of ship peaks that could be analysed at all. This is the distinction that urban-network validation studies systematically obscure: hourly-averaged agreement with a reference monitor says almost nothing about whether a sensor can resolve a transient. Any deployment aimed at source attribution rather than compliance-style mean concentration should be specified on *noise floor and t₉₀*, not on R^2 .

What changed today

- **Five of fourteen records are preprints**, the highest preprint share this watch has carried. All five arrive through the Europe PMC leg (three medRxiv, two Research Square); none has been peer reviewed. They are tiered and summarised on the same terms as the journal records but the tier reflects design, not review status — read the Mandalapu, Walker, Pathak and Fahim entries accordingly.
- **Two independent preprints argue that the exposure metric, not the exposure, is what determines the disparity you observe.** Mandalapu et al. find that 13.21 % of residents of predominantly Black tracts live in joint heat-plus-PM_{2.5} hotspots against 3.33 % in predominantly White tracts, with **OR 3.97 (95 % CI 2.79–5.66)** for HOLC grade D versus grade A after adjustment. Walker et al., measuring directly at 176 Rhode Island sites, find that the social gradient sits in the acoustic environment and night-time light — **short-term PM_{2.5} was roughly an order of magnitude weaker** and showed no reliable gradient. Both are correct; they differ in averaging time and in whether the metric is modelled-annual or measured-instantaneous. Composite environmental-justice indices that mix the two are not measuring one thing.
- **A foundation-model result with a directly usable negative finding.** Fahim & Karim benchmark zero-shot time-series foundation models against tuned specialists across 29 cities. On PM_{2.5} the two are statistically indistinguishable (MASE 0.662 vs 0.692), transfer learning never beats zero-shot at any budget, and under frequent retraining the untrained model uses roughly ten times less measured energy. More useful still: the specialist's apparent temperature advantage (0.533 vs 0.792) collapses to a non-significant 0.745 once its covariates are restricted to values knowable at forecast time — a *perfect-foresight artefact* that will be present in any PM forecasting benchmark that feeds future meteorology to one arm and not the other.
- **mtDNA copy number is not yet a usable PM exposure biomarker.** Pathak et al. pool 24 studies and 13,092 participants and find PM associated with a **−4.90 % (95 % CI −7.97 to −1.82) change in percentage mtDNA-CN** but a **+0.55 (0.05–1.04) change in absolute mtDNA-CN**. The directions disagree because the metrics disagree; that is a measurement-harmonisation failure in the underlying literature, not a biological finding.
- **Indoor combustion, interaction terms, and ventilation.** Chen et al. monitored 24 h indoor PM_{2.5}/PM₁₀ and gases in 60 Taiwanese households and report that PM_{2.5} × TVOC and PM_{2.5} × CO interactions — not PM_{2.5} alone — carried the adverse effects on myocardial contractility and vascular elasticity, with elevated CO₂ acting as an independent stressor in non-incense-burning homes. A PM-only indoor monitor cannot see this.
- **One abstract-free deposit.** Vivanco et al. (*Environ Pollut*) on air-quality-model responsiveness to real-world emission changes arrived with title and DOI but no abstract, and is carried at tier C on the title alone. The Elsevier abstract backlog now spans four weeks.

Corpus at a glance

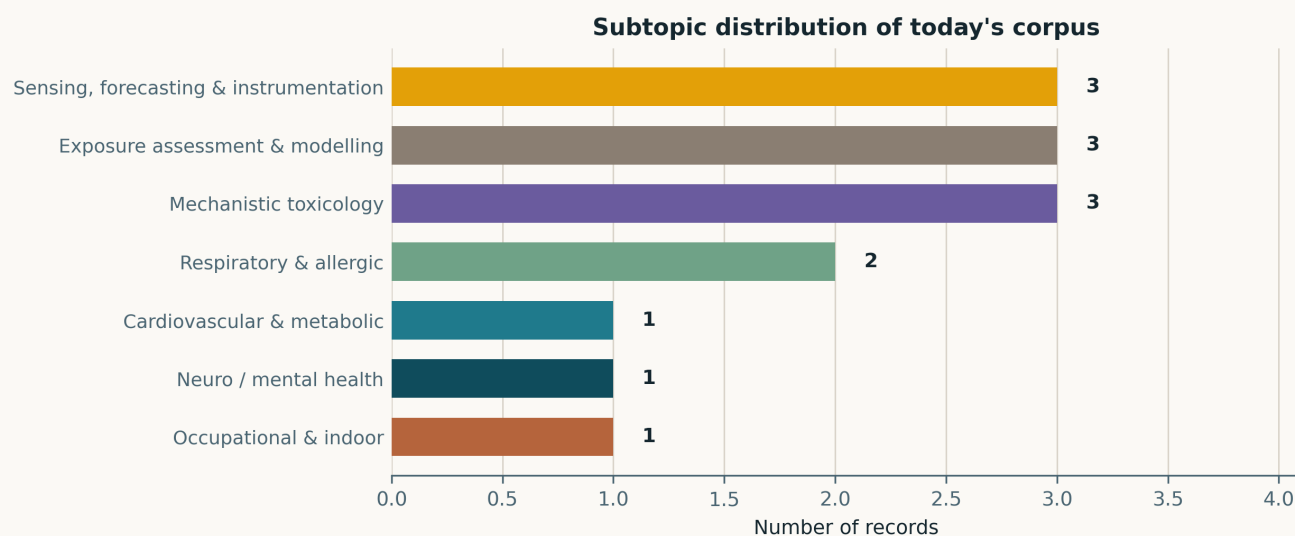


Fig. Records per subtopic cluster. Seven of ten clusters are occupied. *Sensing, forecasting & instrumentation*, *Exposure assessment & modelling* and *Mechanistic toxicology* tie at 3 records each; every mechanistic record is a synthesis rather than a primary experiment. Assignment is single-label by dominant contribution.

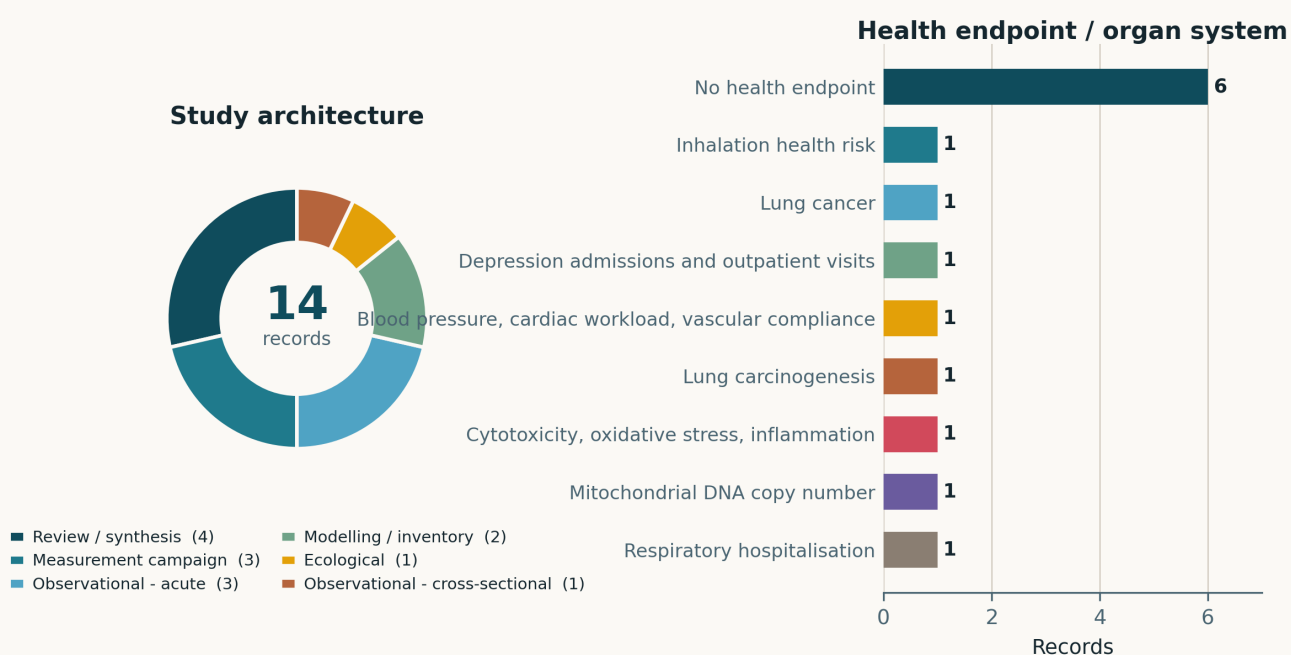


Fig. Left — study architecture as binned by the coarse grouper. *Review / synthesis* leads at 4, with *Measurement campaign* and *Observational — acute* tied at 3; *Modelling / inventory* 2, and one each for *Ecological* and *Observational — cross-sectional*. Note that Eger et al. — a year-long instrument co-location, not an epidemiological field study — is binned as a measurement campaign, the closest available group rather than an accurate description. Right — health endpoint. **Six of the fourteen records carry no human health endpoint at all:** the three sensing/forecasting records and the three exposure-assessment records report concentration, hazard quotient or forecast skill only. The remaining eight endpoints are singletons — no two records in this window share an outcome.

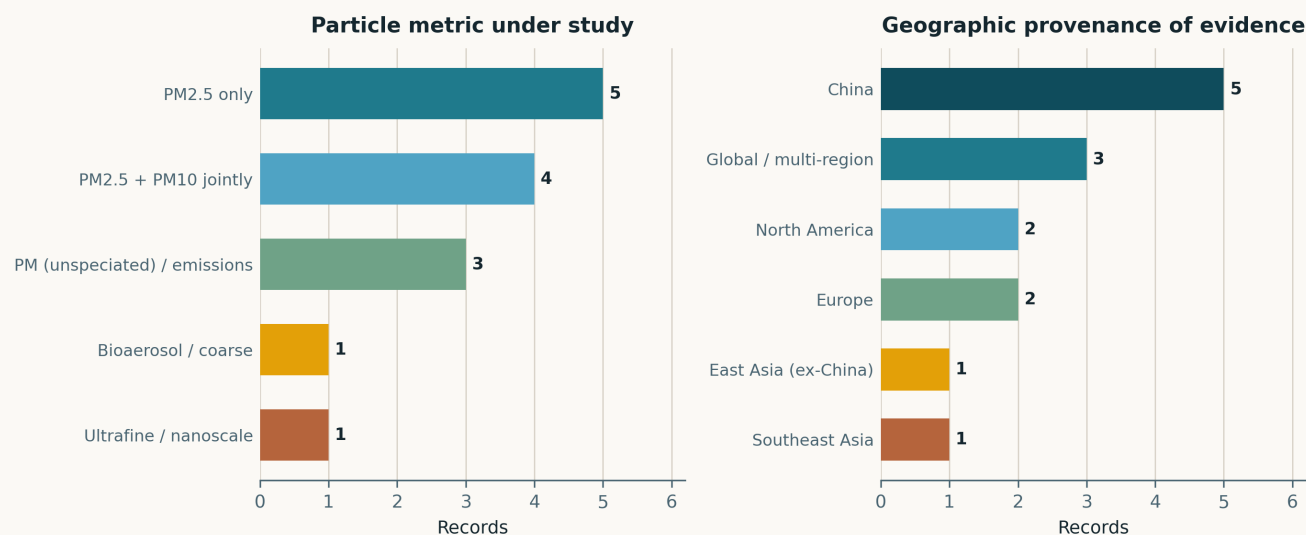


Fig. Left — particle metric. *PM_{2.5} only* 5 and *PM_{2.5} + PM₁₀ jointly* 4 dominate; *PM unspiciated / emissions* 3; *Bioaerosol / coarse* and *Ultrafine / nanoscale* 1 each. The grouper resolved Eger et al. (black carbon plus ultrafine particle number) and Yang et al. (culturable bioaerosol counts) into their own bins rather than folding them into a mass bin. Right — geographic provenance. **China alone supplies 5 of 14 records**; multi-region syntheses 3, North America 2, Europe 2, and one each from East Asia outside China (Taiwan) and Southeast Asia (Thailand). Nothing from Africa, South America or South Asia.

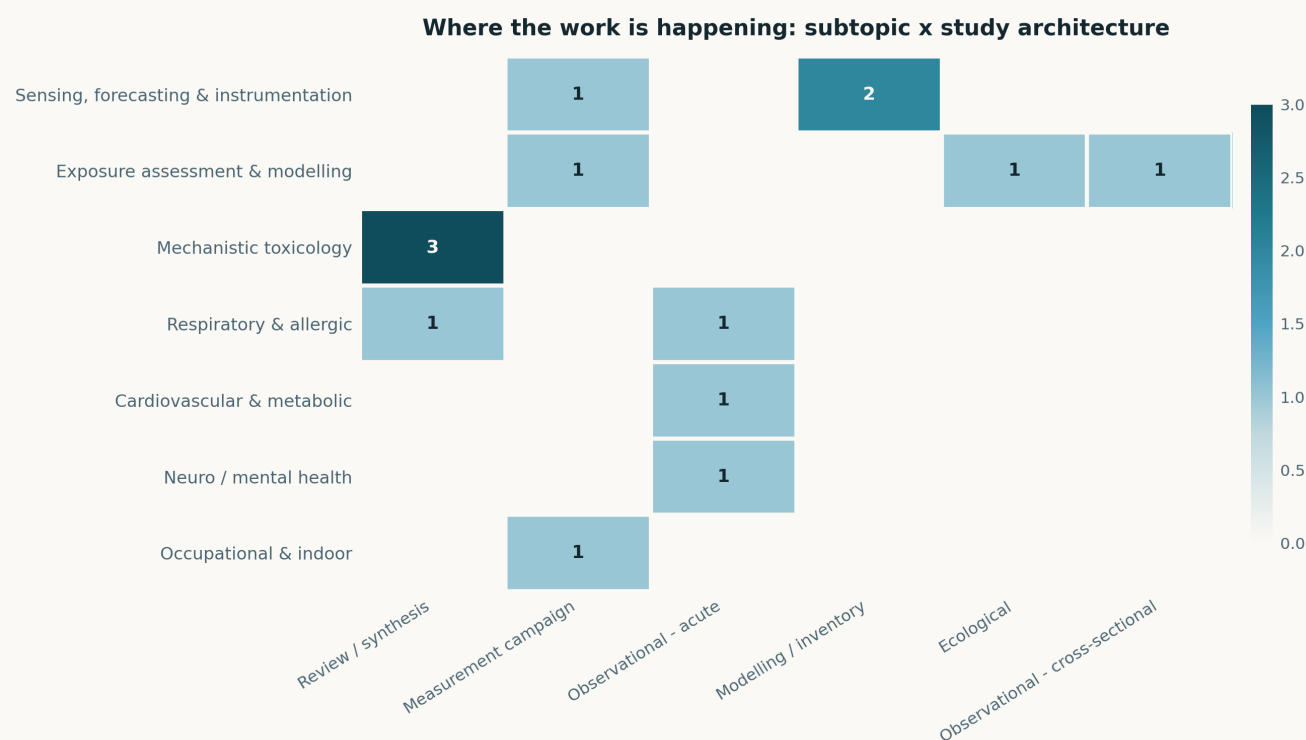


Fig. Subtopic × study-architecture cross-tabulation. The densest cell is *Mechanistic toxicology* × *Review / synthesis* (3) — every mechanistic record in the window is a synthesis, and not one is a primary experiment. The *Sensing* row is split one-and-two across instrument evaluation and modelling, and the *Exposure assessment* row contains no health-endpoint linkage whatsoever.

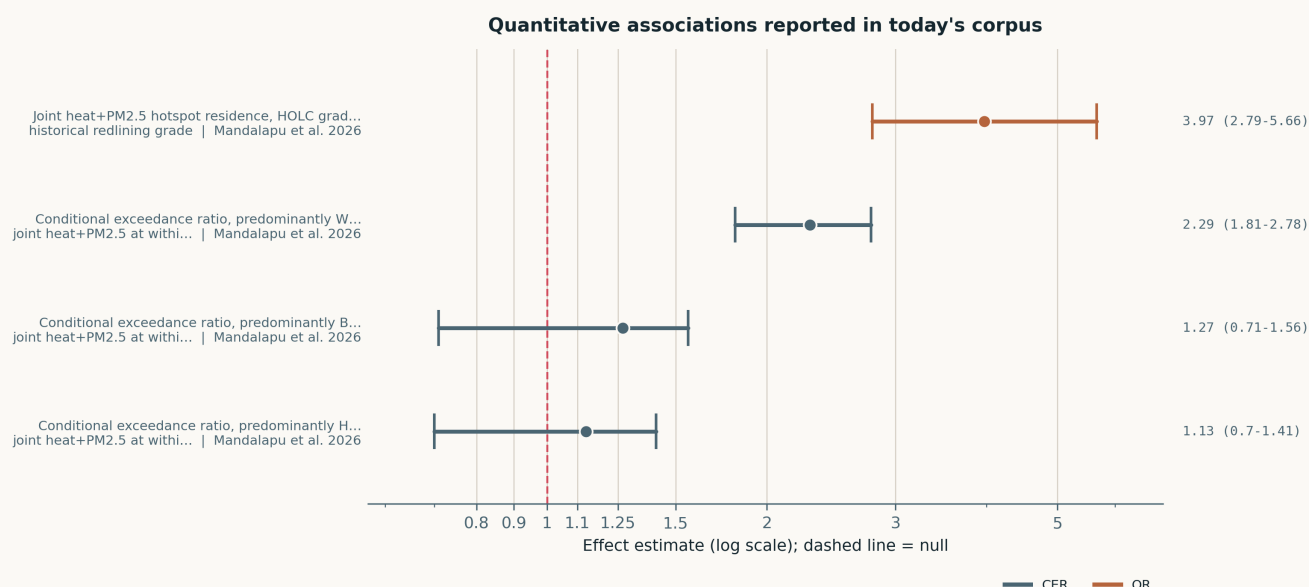


Fig. Every interval-bearing estimate in the window, and **all four come from a single unreviewed preprint** (Mandalapu et al.). The top estimate is an adjusted odds ratio for joint-hotspot residence; the three below it are *conditional exceedance ratios* — observed joint occurrence against within-group statistical independence — and are therefore not on the same scale as the odds ratio and not comparable to it. The Black and Hispanic intervals both include 1; the authors attribute this to limited power from fewer contributing metropolitan areas rather than to an absence of dependence. Read the plot for precision and direction only.

Life-course windows addressed today (records may span several stages)

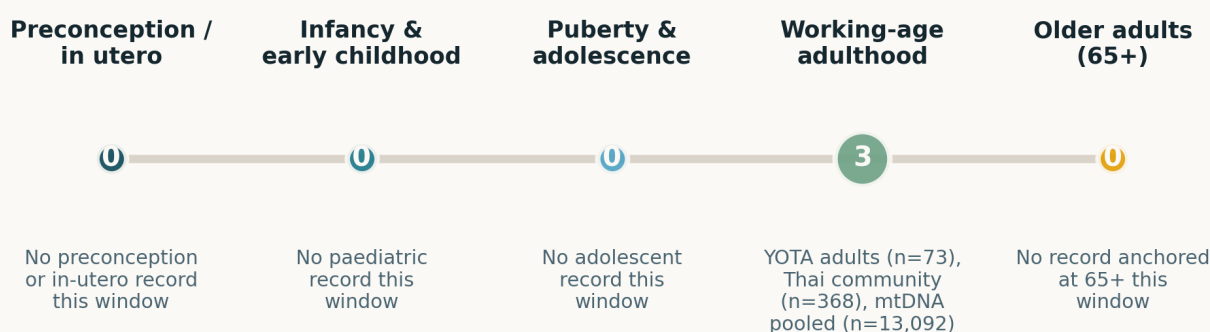


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored to a stage. Three records qualify, all working-age adults. Ten records study no human population at all or pool across ages without stratification; the in-utero, paediatric, adolescent and 65+ windows are empty for the second consecutive issue.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Preprints are labelled as such in the journal field.

Sensing, forecasting & instrumentation

3 records

Eger et al. 2026 Aerosol Air Qual Res

Compact low- and mid-cost sensors for inland-ship exhaust emissions

One year of onshore co-location on the Rhine against reference instruments, screening sensors on noise floor, correlation and response time — criteria chosen because ship plumes are transients, not the slowly-varying urban background most validation studies target. Qualifying devices: **MA350 black carbon** ($R^2 = 0.95$, **slope 0.77**), **Partector 2 Pro ultrafine particle number** ($R^2 = 0.85$), **K30 FR CO₂** ($R^2 = 0.98$), **Alphasense NO** ($R^2 = 0.99$) and **NO₂** ($R^2 = 0.81$). The limitation the authors state plainly is the useful part: sensor noise and the temperature dependence of the electrochemical cells — which they describe as difficult to correct — cut the number of analysable ship peaks. No PM mass sensor cleared the bar, which is itself a result.

doi: 10.1007/s44408-026-00163-w

Fahim & Karim 2026 Res Sq (preprint)

Cost-aware benchmark of time-series foundation models for PM_{2.5} forecasting

Twenty-nine cities, two domains (hourly PM_{2.5}, 2 m temperature), pre-specified protocols with measured energy and a cost-adjusted decision rule, comparing zero-shot foundation models, transfer learning and tuned specialists. **On PM_{2.5} the zero-shot model and the specialist are statistically indistinguishable (MASE 0.662 vs 0.692)**; transfer learning never significantly beats zero-shot at any budget; frequent retraining costs the specialist roughly ten times the measured energy. The methodological contribution is the demonstration that the specialist's temperature advantage is a *perfect-foresight artefact* of feeding it covariates unknowable at forecast time. Unreviewed, and 29 cities is a small sample for a claim about generalisation — but the artefact critique applies to published benchmarks too.

doi: 10.21203/rs.3.rs-10515545/v1

Vivanco et al. 2026 Environ Pollut

Responsiveness of an air quality model to real-world emission changes

Metadata-only deposit — title and DOI without an abstract, so the design, model, domain and emission perturbations are all unknown at the time of writing. The question is a live one (whether CTMs used for planning actually reproduce the concentration response to observed emission changes rather than only to scenario deltas) and the record is carried at tier C so it is not lost. Flagged for abstract retry.

doi: 10.1016/j.envpol.2026.129031

Exposure assessment & modelling

3 records

Mandalapu et al. 2026 medRxiv (preprint)

Joint heat and PM_{2.5} co-exposure across 48 US metropolitan areas

42,304 census tracts, 48 large CBSAs, ~174.6 million residents, summers 2015–2020. The framework combines a composite co-exposure index, a joint exceedance indicator, a conditional exceedance ratio against within-group independence, and an upper-tail dependence parameter estimated both non-parametrically (Capéraà–Fougères–Genest) and with a Gumbel copula, with BCa intervals from a 5,000-replicate block bootstrap. **13.21 % of residents of predominantly Black tracts versus 3.33 % of predominantly White tracts live in joint hotspots; HOLC grade D vs grade A gives OR 3.97 (2.79–5.66)** after adjustment for contemporary racial composition, poverty, renter occupancy and pre-1960 housing. Magnitudes attenuate under near-surface air temperature rather than surface temperature. Unreviewed; the tail-dependence machinery is the transferable contribution.

doi: 10.64898/2026.08.20.26360970

Walker et al. 2026 medRxiv (preprint)

Directly measured noise, PM_{2.5} and night light across Rhode Island

176 sites, 704 site-sessions over roughly a year, spanning the Providence–Central Falls–Pawtucket urban core and four rural municipalities, with A- and C-weighted noise, PM_{2.5}, night-time illuminance and relative humidity carried through an identical model as a negative control. A-weighted noise rose with disadvantage on six of eight area-level markers (**1.3–1.8 dBA per SD; 6.6 dBA between least and most racially diverse neighbourhoods**) and C-weighted levels followed the same gradient, exceeding their A-weighted counterparts at block-group scale for renter occupancy and vehicle absence. **Short-term PM_{2.5} was roughly an order of magnitude weaker**; humidity showed nothing. The negative-control design is what makes the PM null interpretable rather than merely underpowered. Short-term measurement cannot speak to annual-average disparity, and the authors do not claim it does.

doi: 10.64898/2026.08.20.26360933

Mueangsai et al. 2026 Res Sq (preprint)

PM_{2.5} health risk quotient near stone crushing plants, Chonburi

Community cross-sectional study, 368 residents by stratified random sampling, with PM_{2.5} from two fixed stations near a crushing plant averaged and pushed through the USEPA health risk assessment to a hazard quotient. The design's weakness is structural and fatal to individual-level inference: **two fixed monitors supply a single exposure value for every participant**, so all between-person exposure contrast is assumed away and the HQ varies only through the questionnaire covariates. Carried at tier C. The setting — crushing plants generate coarse, mineral-dominated PM that optical sensors size poorly — is the reason it is here at all.

doi: 10.21203/rs.3.rs-10160540/v1

Mechanistic toxicology

3 records

Pathak et al. 2026 medRxiv (preprint)*Airborne PM and mitochondrial DNA copy number: systematic review and meta-analysis*

PRISMA, PROSPERO-registered, JBI risk-of-bias, random-effects pooling; 1,224 records screened to 24 studies and **13,092 participants**, 12 reporting percentage change and 12 absolute values. **Percentage mtDNA-CN fell** (-4.90 ; **95% CI** -7.97 to -1.82 ; $p = 0.002$) **while absolute mtDNA-CN rose** ($+0.55$; **0.05–1.04**; $p = 0.030$). The two directions are not reconcilable as biology; they are an artefact of pooling across incompatible normalisation schemes, and the honest reading is that the underlying literature has not standardised its outcome. Useful as evidence that mtDNA-CN is *not yet* deployable as a PM exposure biomarker, which is the opposite of how it is usually cited.

doi: 10.64898/2026.08.20.26360559

Forest 2026 Sci Total Environ*Molecular mechanisms of sand and dust storm toxicity*

Review of in vitro and in vivo evidence on storm-derived PM₁₀/PM_{2.5}, whose composition — quartz and clay minerals plus adsorbed bacteria, viruses and heavy metals acquired in transport — differs enough from urban PM that the author argues current regulation, which treats storm episodes as generic PM, misses hazard-specific risk. Consistent findings are cytotoxicity, oxidative stress, pro-inflammatory response and exacerbation of bacterial and viral infection. The stated gaps are the substantive content: **most in vivo work uses Asian sand dust while Saharan dust dominates global emissions**, and non-respiratory target organs are barely studied. Directly relevant to any network operating in a dust-influenced region, where the mass a sensor reports and the hazard it implies decouple.

doi: 10.1016/j.scitotenv.2026.182233

Li & Li 2026 Zhongguo Fei Ai Za Zhi*GPER-mediated pulmonary carcinogenesis by fine particulate matter*

Chinese-language narrative review proposing that estrogen-like organic components and metal ions in PM_{2.5} activate the G protein-coupled estrogen receptor, which is overexpressed in lung tumour tissue, driving MAPK/ERK and downstream signalling. No new data and no quantitative synthesis; the receptor-level specificity is the only reason to prefer it to the generic PM-and-lung-cancer literature. Tier C.

doi: 10.3779/j.issn.1009-3419.2026.101.13

Cardiovascular & metabolic

1 record

Chen et al. 2026 Indoor Air*Residential incense burning, indoor pollutants and cardiovascular haemodynamics*

Seventy-three adults from the YOUNG TAIwanese cohort across 60 households, with continuous 24 h indoor PM_{2.5}, PM₁₀, CO, CO₂, TVOC and formaldehyde alongside non-invasive haemodynamic measurement, analysed by mixed-effects models. Incense-burning households showed higher pollutant concentrations and worse haemodynamics — higher blood pressure and cardiac workload, lower vascular compliance. **The finding that matters is interactive: PM_{2.5} × TVOC and PM_{2.5} × CO carried the effects on myocardial contractility and vascular elasticity**, and in non-incense homes elevated CO₂ acted as an independent stressor. No effect sizes are given in the abstract. $n = 73$ with household clustering is thin for interaction terms, and residual confounding by household behaviour is not addressable here. Implication for instrumentation: a PM-only indoor node is blind to the combination that did the work.

doi: 10.1155/ina/8778576

Neuro / mental health

1 record

Ma et al. 2026 Ecotoxicol Environ Saf*Day-on-day increases in air pollution and depression morbidity, 239 Chinese cities*

Time-stratified case-crossover on **330,610 hospital admissions and 3,726,390 outpatient visits** for depression, 239 prefecture-level-or-above cities, 2013–2017. The exposure metric is the novelty: air pollution increase between neighbouring days (APIN), the current day's absolute concentration minus the previous day's, rather than level. Per IQR increase in APIN at lag 0–3, admissions rose between 0.59% (0.06–1.13) for 8 h O₃ and 1.66% (1.09–2.22) for NO₂, with outpatient visits 0.42–0.85%; associations were robust to adjustment for absolute concentrations except for ozone. **PM_{2.5}- and PM₁₀-specific estimates are not reported in the abstract**, only the range endpoints, so no PM estimate is carried into the forest plot. The rate-of-change framing is worth borrowing regardless of the endpoint.

doi: 10.1016/j.ecoenv.2026.120707

Respiratory & allergic

2 records

China Lung Oncology Group 2026 Zhongguo Fei Ai Za Zhi*Regional lung cancer related to coal pollution, Yunnan–Guizhou Plateau*

Multi-sectional, multi-omics regional evidence synthesis proposing “China Wumeng Mountain lung cancer” as a distinct entity driven by long-term household coal combustion, with the argument resting on differences in coal source and mineral composition rather than on total exposure. Clinical and molecular characterisation is paired with a proposed prevention pathway. The construct is at risk of circularity — defining a regional cancer type by its presumed cause and then citing regional incidence as evidence — and the abstract gives no effect estimates. Tier B on the strength of the source-composition argument, which is the same argument the wildfire and dust literatures are making independently.

doi: 10.3779/j.issn.1009-3419.2026.102.16

Yang et al. 2026a GeoHealth*Spatiotemporal risk of environmental drivers of respiratory disease, Jiangsu*

Distributed lag non-linear models coupled to geographically and temporally weighted regression on respiratory hospitalisations across Jiangsu Province, 2020–2023, a population containing over 27 million children under 15 and adults over 65. The DLNM handles the lag structure and the GTWR lets the exposure–response coefficient vary in space and time rather than assuming one province-wide slope — which is the right question for a province with this much industrial heterogeneity. The 2020–2023 window overlaps pandemic-era mobility and healthcare-seeking disruption; the abstract does not say how that is handled, and it is the main threat to the temporal component.

doi: 10.1029/2026gh001881

Occupational & indoor

1 record

Yang et al. 2026b Environ Pollut*Bioaerosols at oil exploitation sites and mask filtration efficiency*

Seasonal sampling in a Beijing–Tianjin–Hebei oil city carrying both heavy PM and VOC loads, characterising culturable and total bioaerosol distribution, sources and inhalation risk, plus mask filtration efficiency. **Autumn levels significantly exceeded summer while microbial activity moved the other way** — a dissociation between count and viability that matters for anyone interpreting a fluorescence channel as a proxy for biological hazard. Culture-based enumeration undercounts viable-but-non-culturable organisms by a large and site-dependent factor, which is the standing limitation of this entire measurement class. Tier C.

doi: 10.1016/j.envpol.2026.129020

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Ma et al.	Associations between short-term absolute increases in ambient air pollution and hospital admissions and outpatient visits for depression: A nationwide case-crossover study	<i>Ecotoxicol Environ Saf</i>	Time-stratified case-crossover	Day-on-day absolute increase (APIN) in PM _{2.5} and PM ₁₀	China
Cardiovascular & metabolic					
Chen et al.	Residential Incense Burning, Indoor Air Pollution, and Cardiovascular Hemodynamic Alterations	<i>Indoor Air</i>	Repeated-measures panel, 24 h indoor monitoring	24 h indoor PM _{2.5} and PM ₁₀ with CO, CO ₂ , TVOC, HCHO	Taiwan
Respiratory & allergic					
China Lung Oncology Group	China Wumeng Mountain Lung Cancer: Evidence Chain, Clinical and Molecular Characteristics of Regional Lung Cancer Related to Coal Pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Multi-omics regional evidence synthesis	Household coal-combustion PM, source-resolved mineral composition	China
Yang et al.	Analysis of Spatiotemporal Risks of Environmental Driving Factors of Respiratory Diseases	<i>GeoHealth</i>	DLNM + geographically-temporally weighted regression	Ambient PM and co-pollutants, prefecture daily	China
Mechanistic toxicology					
Forest	Toxicological impact of sand and dust storms on human health: A review on the molecular mechanisms involved	<i>Sci Total Environ</i>	Mechanistic review	Storm-derived PM ₁₀ and PM _{2.5} (mineral, biological, metal fractions)	Multi-region
Li & Li	Research Advances in the Pulmonary Carcinogenic Effects of Atmospheric Fine Particulate Matter Mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Mechanistic review	PM _{2.5} estrogen-like components and metal ions	China
Pathak et al.	Effect of airborne particulate matter on mtDNA copy number: A systematic review and meta-analysis	<i>medRxiv (preprint)</i>	Systematic review & meta-analysis	Ambient PM _{2.5} / PM ₁₀ , study-specific contrasts	Multi-region
Occupational & indoor					
Yang et al.	Bioaerosols in the atmospheric environment of oil exploitation sites: spatial and temporal distribution, sources, influencing factors, health risks, and reduction efficiency of masks	<i>Environ Pollut</i>	Seasonal field sampling + filtration test	Culturable and total bioaerosol, particle-bound microorganisms	China
Sensing, forecasting & instrumentation					
Eger et al.	Applicability of Compact Low- and Mid-Cost Sensors to Monitor Air Pollutant Emissions from Inland Ships	<i>Aerosol Air Qual Res</i>	Year-long co-location field evaluation	BC, ultrafine particle number, NO/NO ₂ , CO ₂ in ship plumes	Germany
Fahim & Karim	Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting	<i>Res Sq (preprint)</i>	Benchmark / forecasting model comparison	Hourly PM _{2.5} concentration	Multi-country (29 cities)
Vivanco et al.	Assessing the responsiveness of an air quality model to real-world emission changes for air quality planning	<i>Environ Pollut</i>	Chemical transport model evaluation	Modelled PM and gaseous pollutant response to emission perturbation	Europe
Exposure assessment & modelling					
Mandalapu et al.	Joint Heat and PM _{2.5} Exposure Across US Metropolitan Areas: Multi-Stressor Disparities, Historical Redlining, and a Multi-Metric Assessment Framework	<i>medRxiv (preprint)</i>	Ecological cross-sectional, copula tail dependence	Summer mean PM _{2.5} , tract-level, jointly with surface heat	United States
Mueangsai et al.	Factors Associated with Health Risk Quotient among Communities Exposed to PM _{2.5} from Stone Crushing Plants in Chonburi Province Thailand	<i>Res Sq (preprint)</i>	Community cross-sectional + USEPA HQ	Fixed-site PM _{2.5} , two stations, community average	Thailand
Walker et al.	Divergent social patterning of directly measured environmental exposures across Rhode Island communities	<i>medRxiv (preprint)</i>	Direct-measurement field campaign	Short-term PM _{2.5} alongside LAeq/LCeq noise and nighttime illuminance	United States

Provenance and method

Entry window **2026-08-25 to 2026-08-25**, a single complete day of deposits. **Sources.** PubMed E-utilities on the [Date - Entry] field, run as two separate queries (health axis 9 hits, instrumentation axis 0); the PubMed connector returned 13 on the health axis and 0 on the instrumentation axis. Europe PMC CREATION_DATE 13 hits. Crossref by-ISSN across the 24 tracked non-PubMed-indexed titles, 43 deposits after front-matter suppression. OpenAlex returned empty for the 23rd consecutive run — the date-filter leg has been broken since July and is not being repaired. arXiv contributed nothing new. **Screening.** 74 raw records, 65 unique after PMID/DOI/title collapse, 2 already present in `seen.json`, **14 retained in scope and 51 rejected** with reason logged. Rejections were dominated by non-particle atmospheric chemistry (ozone, VOC, ammonia, gravity waves), non-inhalation ecotoxicology, and eleven Elsevier retraction notices. **Preprints.** Five records are unreviewed preprints (medRxiv, Research Square). Tier reflects study design only and carries no statement about review status. **Effect estimates.** All four interval-bearing estimates come from one preprint and mix an odds ratio with three conditional exceedance ratios; they are plotted on a common axis for precision and direction and are not mutually comparable. Estimates from Pathak et al. are mean differences on an unbounded scale and are reported in the digest text rather than plotted. **Labels.** Subtopic, design, particle-metric and geography are assigned from title, abstract and keywords; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. All DOI links were resolved before build.